LAB FILE

C++ Programming

Lab 5

Loops | Patterns | Control Statements

```
for (int i = 1; i <= 7; i++)
        cout << "*";
```


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7 programs | code | output | handwritten pages | think questions

Contents

Everything in this file, in order

1 Task 1: Print the table of any number — Page 3

2 Task 2: Find the power of a number — Page 4

3 Task 3: Sum of natural numbers using a loop — Page 5

4 Task 4: Sum of digits of a number — Page 6

5 Task 5: Factorial of a number — Page 7

6 Task 6: Print a star pattern — Page 8

7 Task 7: Days in a month using switch — Page 9

HW Handwritten pages (8 pages) — Pages 10 - 17

Each task page shows the typed program with its output. The handwritten copy of the same work is at the end.

TASK 1

Print the table of any number

C++ program | for loop, cin / cout

Reads a number and prints its multiplication table (1 to 10) using a for loop.

PROGRAM

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num;

    cout << "Enter any number: ";
    cin >> num;

    for (int i = 1; i <= 10; i++)
    {
        cout << num << " * " << i << " = " << num * i << endl;
    }

    return 0;
}
```

OUTPUT

```
Enter any number: 12
12 * 1 = 12
12 * 2 = 24
12 * 3 = 36
12 * 4 = 48
12 * 5 = 60
12 * 6 = 72
12 * 7 = 84
12 * 8 = 96
12 * 9 = 108
12 * 10 = 120

[Program finished]
```

TASK 2

Find the power of a number

C++ program | for loop, accumulator

Multiplies the number by itself "power" times using a for loop. Example: 2 to the power 3 gives 8.

PROGRAM

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num, power, result = 1;

    cout << "Enter the number: ";
    cin >> num;

    cout << "Enter the power: ";
    cin >> power;

    for (int i = 1; i <= power; i++)
    {
        result = result * num;
    }

    cout << "Result = " << result;

    return 0;
}
```

OUTPUT

```
Enter the number: 2
Enter the power: 3
Result = 8
[Program finished]
```

TASK 3

Sum of natural numbers using a loop

C++ program | for loop, running sum

Adds 1 + 2 + ... + n using a for loop. Example: n = 5 gives 15.

PROGRAM

```cpp
#include <iostream>
using namespace std;

int main()
{
    int n, sum = 0;

    cout << "Enter a number: ";
    cin >> n;

    for (int i = 1; i <= n; i++)
    {
        sum = sum + i;
    }

    cout << "Sum of natural numbers = " << sum;

    return 0;
}
```

OUTPUT

```
Enter a number: 5
Sum of natural numbers = 15
[Program finished]
```

TASK 4

Sum of digits of a number

C++ program | while loop, % and / operators

Takes the last digit with % 10, adds it, then removes it with / 10 inside a while loop. Example: 358 gives 16.

PROGRAM

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num, digit, sum = 0;

    cout << "Enter a number: ";
    cin >> num;

    while (num > 0)
    {
        digit = num % 10;
        sum = sum + digit;
        num = num / 10;
    }

    cout << "Sum of digits = " << sum;

    return 0;
}
```

OUTPUT

```
Enter a number: 358
Sum of digits = 16
[Program finished]
```

TASK 5

Factorial of a number

C++ program | for loop, product

Multiplies all numbers from 1 to n using a for loop. Example: 4! = 24.

PROGRAM

```cpp
#include <iostream>
using namespace std;

int main()
{
    int num;
    int fact = 1;

    cout << "Enter a number: ";
    cin >> num;

    for (int i = 1; i <= num; i++)
    {
        fact = fact * i;
    }

    cout << "Factorial = " << fact;

    return 0;
}
```

OUTPUT

```
Enter a number: 4
Factorial = 24
[Program finished]
```

TASK 6

Print a star pattern

C++ program | nested for loops

Nested for loops print a right-angled triangle of stars (7 rows).

PROGRAM

```cpp
#include <iostream>
using namespace std;

int main()
{
    for (int i = 1; i <= 7; i++)
    {
        for (int j = 1; j <= i; j++)
        {
            cout << "*";
        }
        cout << endl;
    }

    return 0;
}
```

OUTPUT

```
*
**
***
****
*****
******
*******

[Program finished]
```

TASK 7

Days in a month using switch

C++ program | switch, case, break, default

A switch statement groups months into 31-day, 30-day and February cases. Input 2 shows "28 or 29 days".

PROGRAM

```c
#include <stdio.h>

int main()
{
    int month;

    printf("Enter month number (1-12): ");
    scanf("%d", &month);

    switch (month)
    {
        case 1:
        case 3:
        case 5:
        case 7:
        case 8:
        case 10:
        case 12:
            printf("31 days");
            break;

        case 4:
        case 6:
        case 9:
        case 11:
            printf("30 days");
            break;

        case 2:
            printf("28 or 29 days");
            break;

        default:
            printf("Invalid month number");
    }

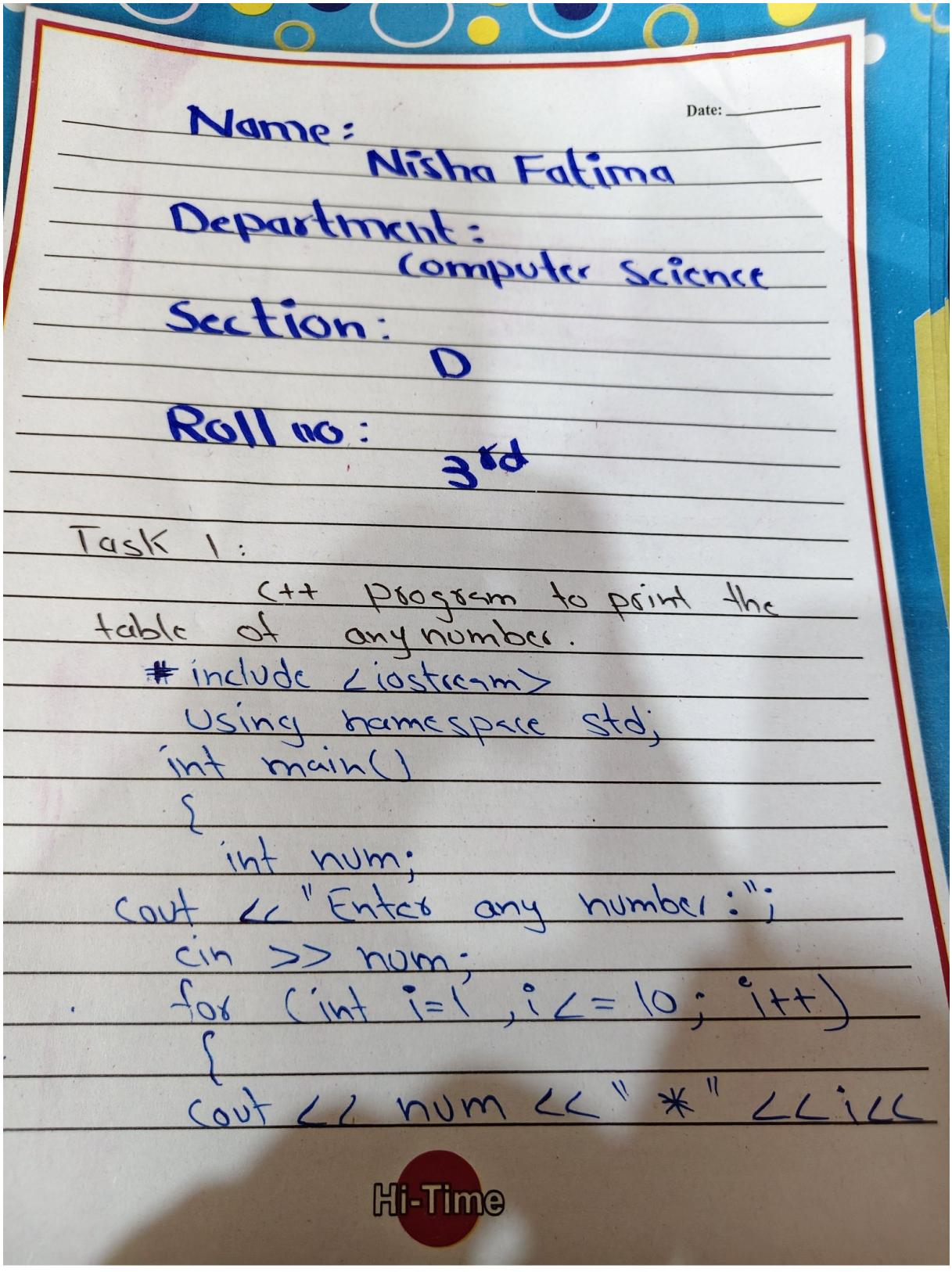
    return 0;
}
```

OUTPUT

```
Enter month number (1-12): 2
28 or 29 days
[Program finished]
```

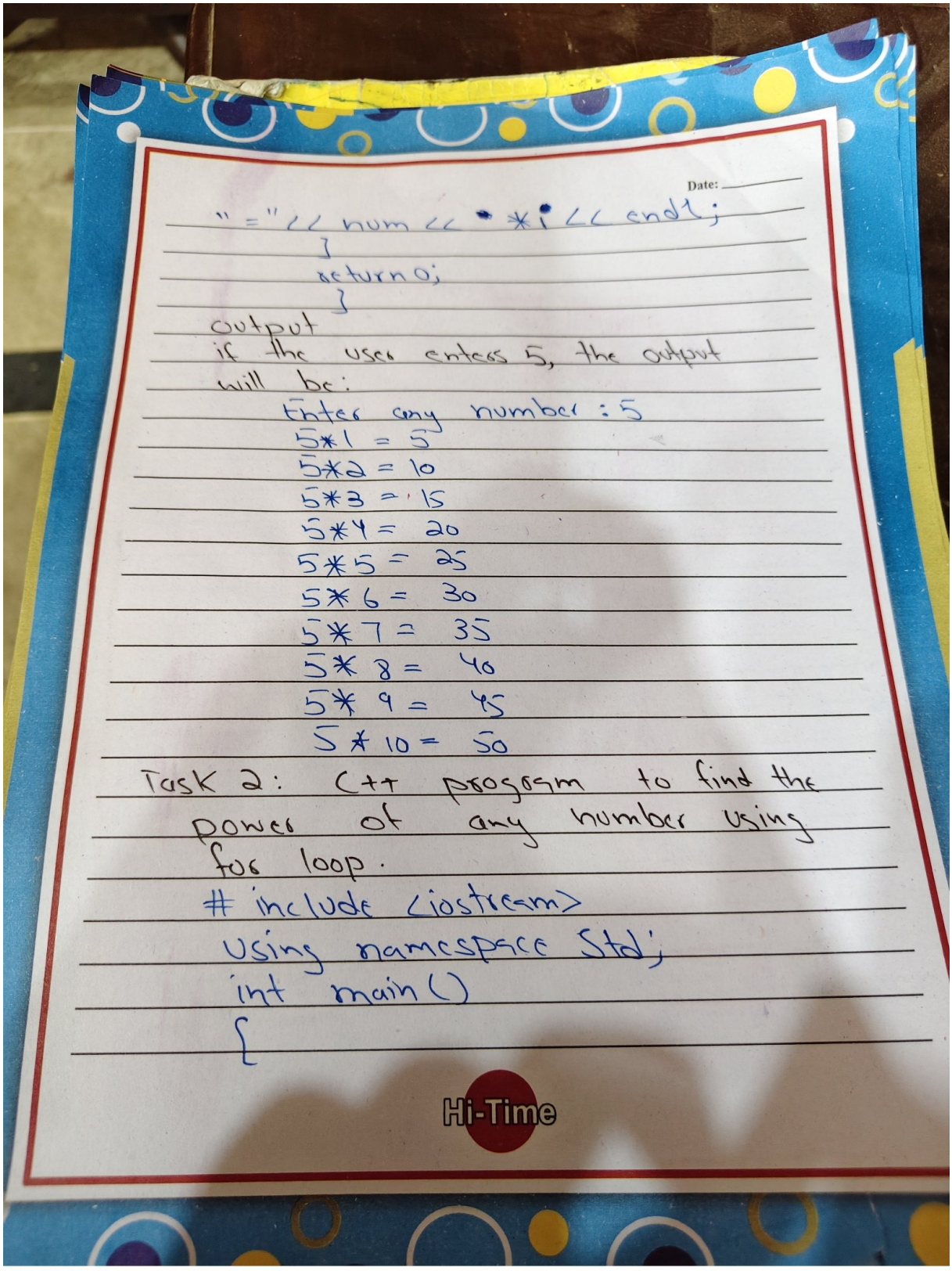
Handwritten Page 1 of 8

Name, department, section, roll no. and Task 1 (table)



Handwritten Page 2 of 8

Task 1 (end) and output, Task 2 (power) begins



" = " << num << " * " << i << endl;
}
return 0;
}

output
if the user enters 5, the output will be:

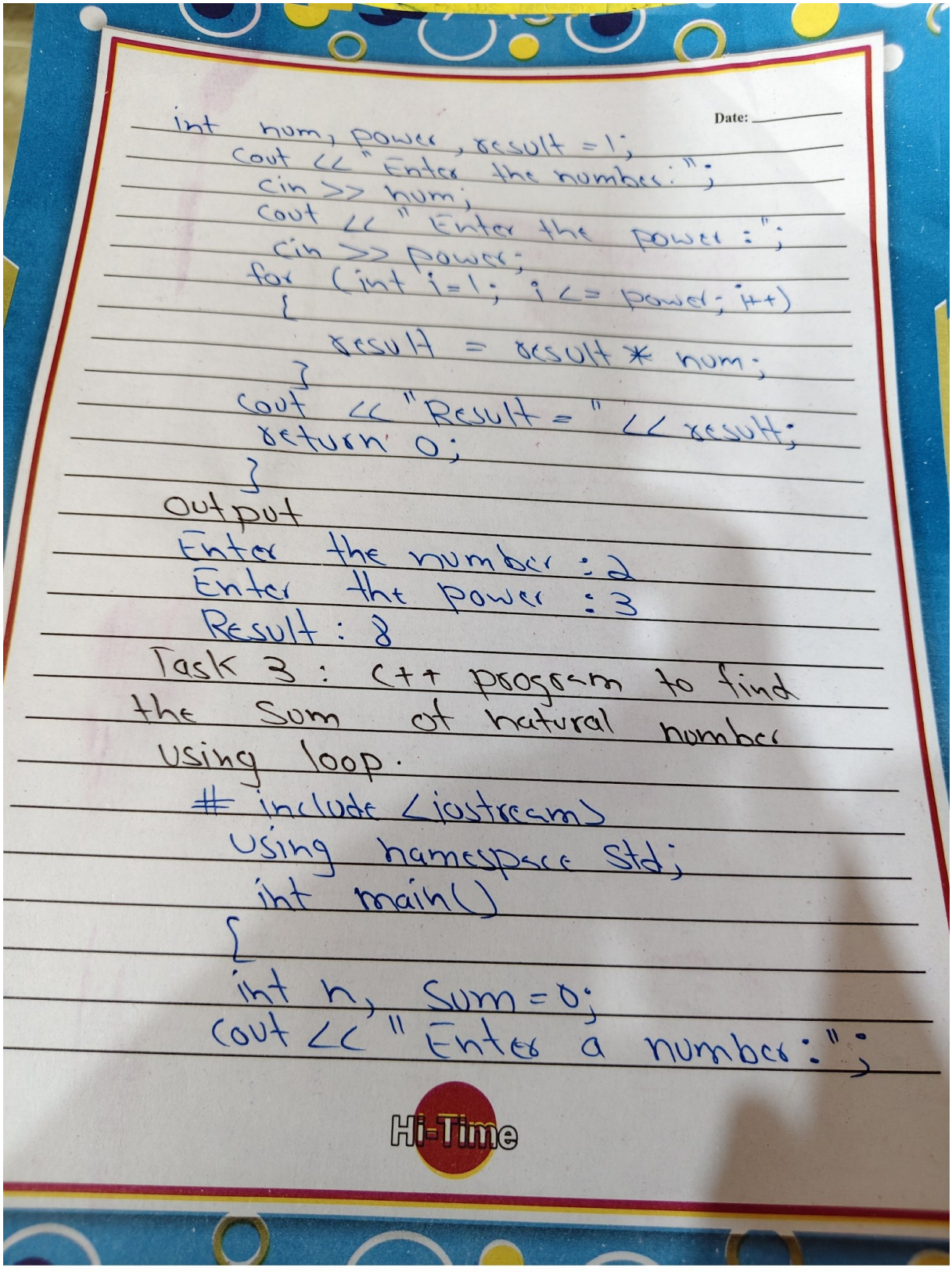
Enter any number : 5
5*1 = 5
5*2 = 10
5*3 = 15
5*4 = 20
5*5 = 25
5*6 = 30
5*7 = 35
5*8 = 40
5*9 = 45
5*10 = 50

Task 2: C++ program to find the power of any number using for loop.

```
#include <iostream>
using namespace std;
int main()
{
```

Handwritten Page 3 of 8

Task 2 (end) and output, Task 3 (sum of natural numbers) begins



Date: _____

```
int num, power, result = 1;
cout << "Enter the number: ";
cin >> num;
cout << "Enter the power: ";
cin >> power;
for (int i=1; i<= power; i++)
{
    result = result * num;
}
cout << "Result = " << result;
return 0;
}
```

Output

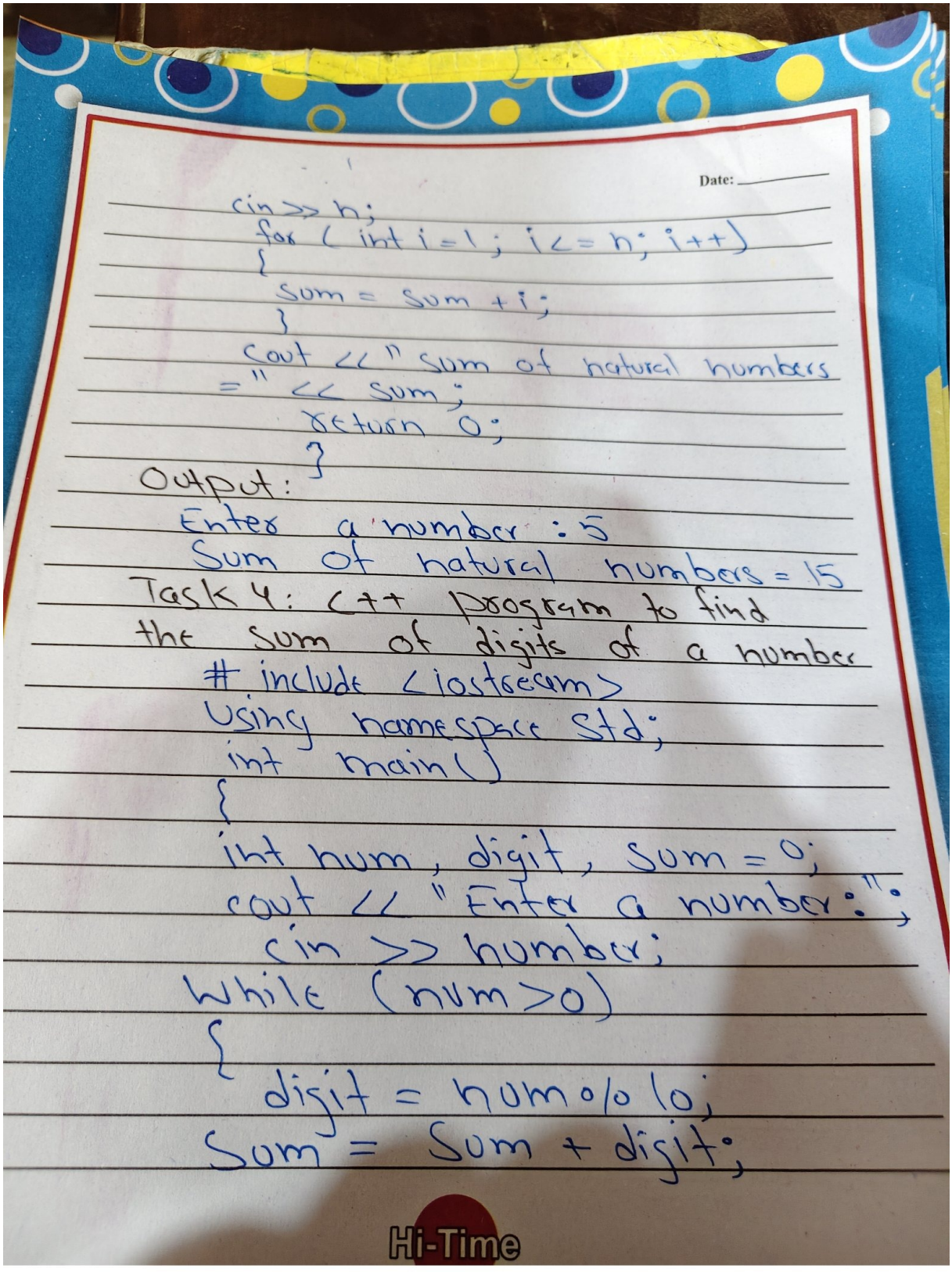
Enter the number : 2
Enter the power : 3
Result : 8

Task 3 : C++ program to find the sum of natural number using loop.

```
#include <iostream>
using namespace std;
int main()
{
int n, sum = 0;
cout << "Enter a number: ";
```

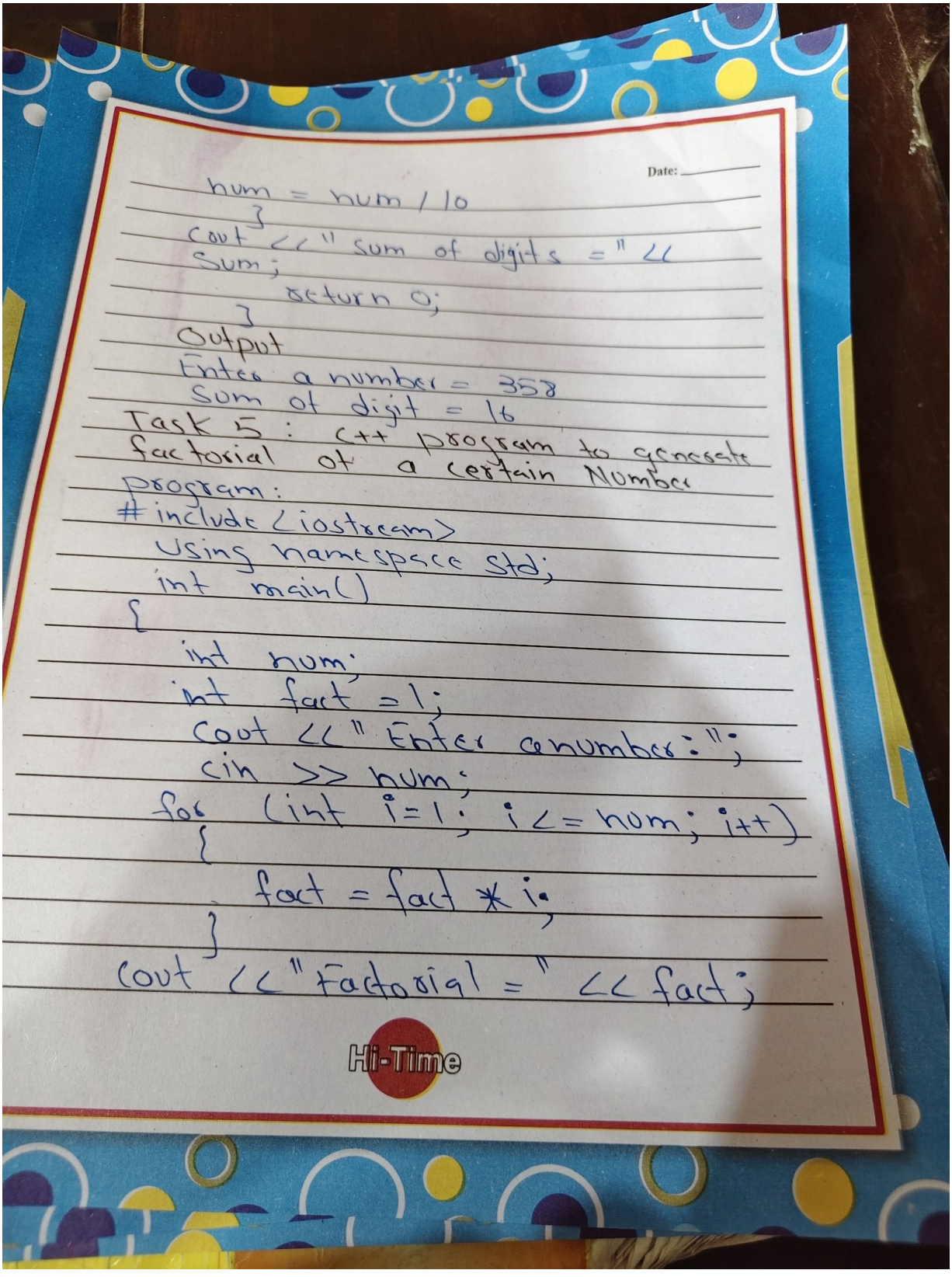
Handwritten Page 4 of 8

Task 3 (end) and output, Task 4 (sum of digits) begins



Handwritten Page 5 of 8

Task 4 (end) and output, Task 5 (factorial) begins



Date:

```
    num = num / 10
    }
cout << " sum of digits = " <<
sum;
    return 0;
    }
```

Output

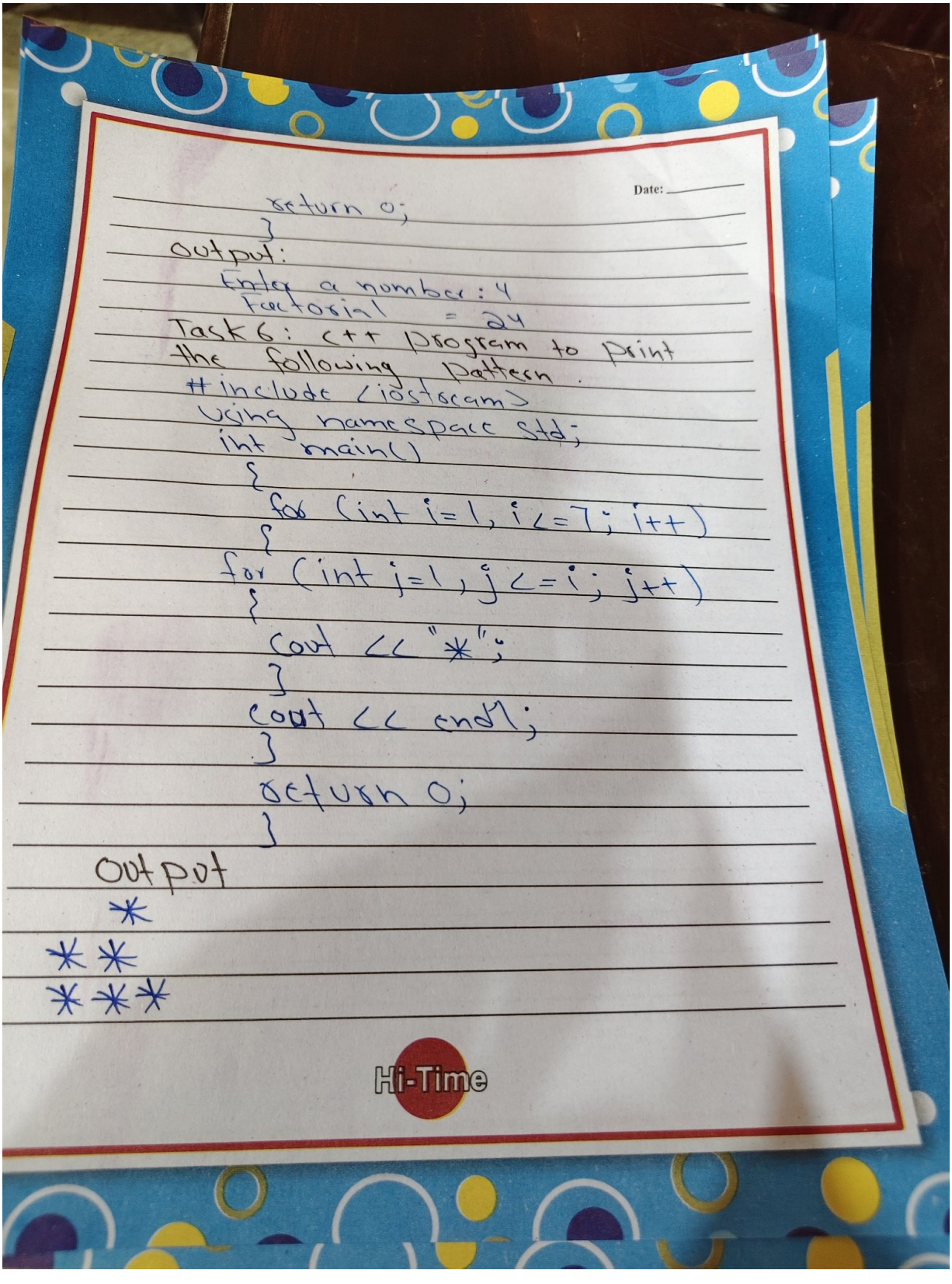
Enter a number = 358
Sum of digit = 16

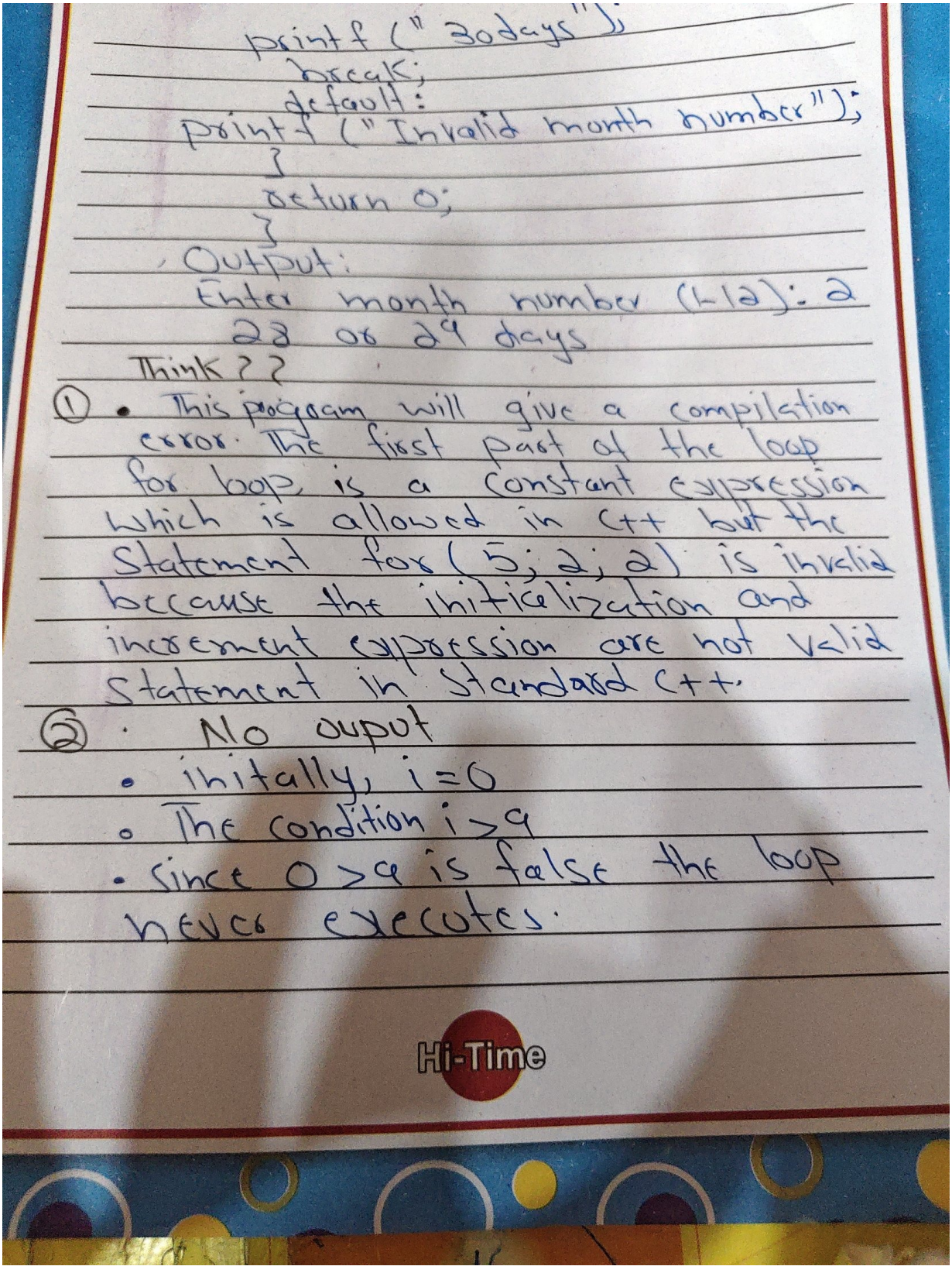
Task 5 : c++ program to generate factorial of a certain Number

program:

```
#include <iostream>
using namespace std;
int main()
{
    int num;
    int fact = 1;
    cout << " Enter a number: ";
    cin >> num;
for (int i=1; i<=num; i++)
    {
        fact = fact * i;
    }
cout << "Factorial = " << fact;
```

Task 5 (end) and output, Task 6 (star pattern)





```
        printf (" 30days");
        break;
        default:
    printf ("Invalid month number");
        }
        return 0;
        }
```

Output:

Enter month number (1-12): 2
28 or 29 days

Think??

① • This program will give a compilation error. The first part of the loop for loop, is a constant expression which is allowed in C++ but the statement for(5; 2; 2) is invalid because the initialization and increment expression are not valid statement in Standard C++.

② • No ouput
• initally, i=0
• The condition i>9
• since 0>9 is false the loop never executes.

Handwritten Page 8 of 8

Think?? answers 3 and 4

